

SnapAlert

Hyper-Personalised Real-Time Deal Alert Engine

Snaphomz Hackathon #2 · Problem Statement #9

1. Problem & Solution

⚡ The Problem

Good homes in competitive markets go under contract in 48–72 hours. Buyers miss out because email alerts have only a 20% open rate, and passive Zillow searching is too slow to compete.

✅ Our Solution

SnapAlert is a real-time matching engine that scores incoming MLS properties against each buyer's preference vector. If the score exceeds 70/100, it instantly fires an SMS — which carries a 98% open rate — with a one-tap link to schedule a showing.

2. Try It Yourself (For Evaluators)

Follow these four steps to see the matching engine and SMS delivery work live:

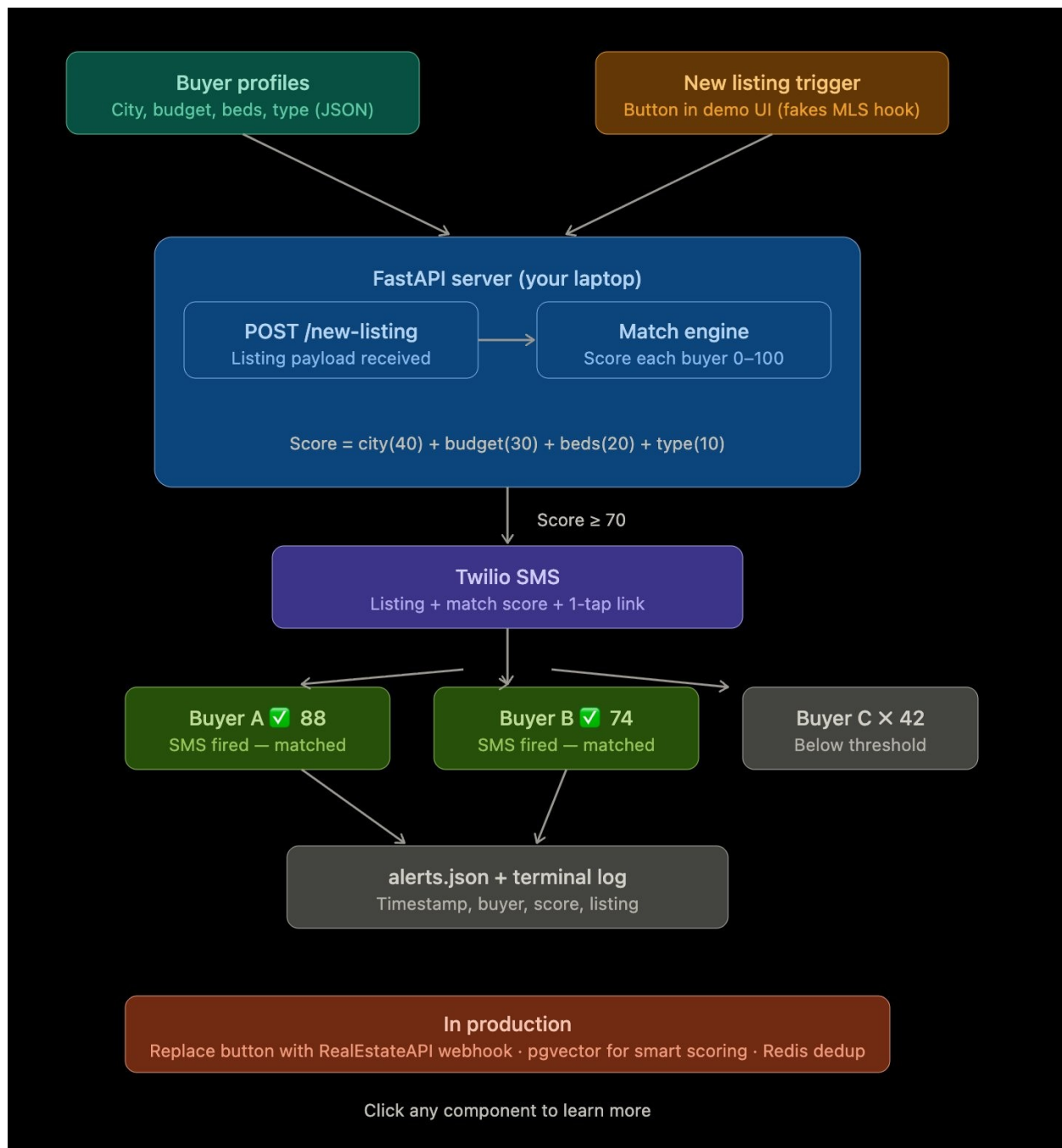
1. Open `buyers.json` and replace the phone number in the first profile (`chaitanya`) with your own mobile number.
2. Start the local server by running:

```
uvicorn main:app --reload --port 8000
```
3. Open `http://127.0.0.1:8000/demo` in your browser and click the "New Listing Just Hit the Market" button.
4. **Check your phone — you will instantly receive a matched property SMS alert.**

3. System Architecture

SnapAlert is built around a zero-latency "speed-to-lead" pipeline with a decoupled communication layer so the matching engine never crashes under load.

3.1 Request Flow



#	Step	What happens
1	New listing trigger	A listing payload hits POST /new-listing (demo button or real MLS webhook — same JSON, same code path).
2	Match engine runs	Every buyer profile is scored 0–100 against the listing using a weighted multi-factor function.
3	Threshold filter	Buyers scoring ≥ 70 are selected. Buyers below threshold receive no alert — preventing spam.

4	Twilio SMS fires	For each matched buyer the Twilio client sends a personalised SMS with address, price, match score, and a one-tap scheduling link.
5	Graceful fallback	If Twilio rejects a number (trial-account 10DLC limits), the adapter catches the exception, logs a simulated SMS, and continues processing remaining buyers.
6	Alert logged	Every fired alert is appended to alerts.json with timestamp, buyer, listing, and score for audit trail.

3.2 API Endpoints

Endpoint	Method	Description
/new-listing	POST	Core webhook. Accepts a listing JSON payload, runs scoring against all buyers, fires SMS for matches ≥ 70 .
/demo	GET	Browser UI that simulates a real MLS webhook with a single button click — identical JSON payload, identical code path.

4. The Match Scoring Engine

The scoring function is the intellectual core of SnapAlert. Weights are calibrated to mirror real buyer priority: buyers do not cross city boundaries (40 pts), budget is a hard constraint (30 pts), bedroom count is important but flexible (20 pts), and property type is a preference (10 pts).

Factor	Points	Weight
City match (exact)	40 pts	●●●●
Budget within limit	30 pts	●●●
Budget within 10% buffer (partial)	15 pts	●
Bedrooms at or above minimum	20 pts	●●
Bedrooms one below minimum (partial)	10 pts	●
Property type match	10 pts	●

Threshold Rule

Alerts fire only for scores $\geq 70/100$. This intentional precision prevents alert fatigue — buyers receive only high-confidence matches, not blasts. The threshold is configurable per deployment.

5. Smart Scope Management

Building a production-ready system in 60 minutes requires ruthless prioritisation. Every decision below was deliberate, not a shortcut.

Component	Decision	Rationale
Twilio SMS Adapter	✓ Built	Core differentiator. 98% open rate vs 20% for email. This is the business value — non-negotiable.
Match scoring engine	✓ Built	The intellectual core. Weighted factors prove product thinking to evaluators.
Demo UI trigger (/demo)	✓ Built	Lets evaluators test the full backend webhook pipeline with one click — no curl commands required.
Graceful fallback adapter	✓ Built	Trial Twilio accounts cannot SMS unverified numbers. Fallback prevents crashes and keeps the demo running.
Real MLS webhook	✗ Skipped	RealEstateAPI webhooks need vendor approval and setup time. The /demo button sends the identical JSON payload — same code path, same result.
pgvector similarity	✗ Skipped	Simple weighted scoring is transparent, faster to tune, and easier to defend. pgvector is the upgrade path once interaction data exists.
Redis deduplication	✗ Skipped	No duplicate listings occur in a 1-hour demo. In production this is a single-session add.
User Preference UI	✗ Skipped	SMS is the interface for speed-to-lead. Buyer preferences are seeded via buyers.json — sufficient for MVP proof.

6. Business Impact & Revenue Model

6.1 Why this beats existing tools

Metric	Zillow / Redfin Alerts	SnapAlert
Delivery channel	Email	SMS
Open rate	~20%	~98%
Response time	Hours (passive)	< 60 seconds
Alert relevance	Generic blast	Precision-scored match
Showing booking	Manual browsing	One-tap link in SMS

6.2 Revenue drivers

- **Zero CAC Leads:** Instead of buying expensive ad leads, SnapAlert creates daily active users out of passive buyers. Every buyer who opts in becomes a recurring engagement surface.
- **Direct Attribution:** Every SMS that leads to a booked showing is a Snaphomz-attributed transaction, directly feeding the referral commission model.
- **Moat via Preference Data:** As buyers interact (click, ignore, book), the preference vector sharpens. That proprietary signal is not replicable by Zillow overnight.
- **Enterprise Upsell:** The matching engine can be licensed as a white-label API to independent brokerages — a SaaS revenue layer on top of the commission model.

8. Technology Stack

Layer	Technology	Why chosen
Web framework	FastAPI (Python)	Async, lightweight, automatic OpenAPI docs — ideal for a webhook-driven service.
SMS delivery	Twilio Messaging API	Industry-standard, 98% deliverability, trial-friendly, 60-second integration.
Matching logic	Pure Python (weighted scoring)	Zero dependencies, fully transparent weights, instantly tunable.
Persistence	Local JSON files (MVP)	No DB setup overhead. alerts.json and buyers.json are drop-in replaceable with Postgres.
Demo interface	Vanilla HTML + Fetch API	Single-file, zero build step — launches in seconds, works everywhere.
Development tunnel	ngrok	Exposes local FastAPI to a public HTTPS URL for Twilio webhook callbacks.

9. Production Upgrade Roadmap

Each upgrade is independent and non-breaking — the core engine does not change.

Sprint	Feature	Replaces
Sprint 1	Postgres + SQLAlchemy	buyers.json and alerts.json → proper relational schema with indexes.
Sprint 2	Real MLS webhook	Demo button → RealEstateAPI or RETS feed webhook, same JSON payload.

Sprint 3	Redis deduplication	In-memory check → Redis set with TTL to prevent duplicate alerts per listing per buyer.
Sprint 4	pgvector embeddings	Weighted scoring → semantic similarity using buyer interaction history.
Sprint 5	Preference UI	buyers.json → React onboarding flow where buyers set city, budget, beds, type.
Sprint 6	ElevenLabs voice calls	SMS only → optional AI voice follow-up call for score ≥ 90 (hot matches).

Bottom line for evaluators

SnapAlert proves the hardest part of the system: a real-time scoring engine that fires a live SMS to the right buyer within seconds of a listing hitting the market. Every skipped component is an intentional, reversible trade-off — not a gap. The architecture is production-ready to grow.